Week 1 Tutorial

Welcome to Week 1 Tutorial. Please follow these instructions step by step - your tutors will also take you through these one by one in class. Please ask for their assistance if you're stuck at any place - we are here to help. Today, you will learn the following:

1: Install **Visual Studio Code (https://code.visualstudio.com/)**

2: Install **Live Server (https://marketplace.visualstudio.com/items?itemName=ritwickdey.LiveServer)** plugin in Visual Studio Code to run a local web server to serve your webpages

3: Introduction to HTML

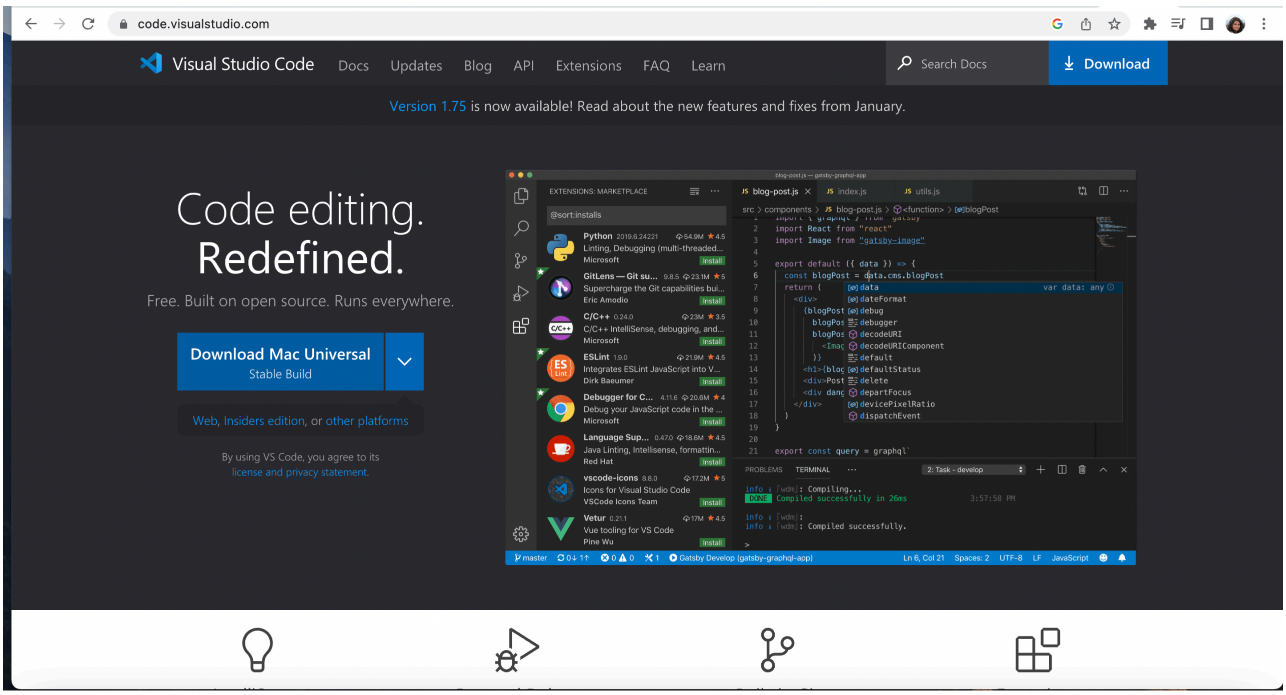
4: Understand the Developer Tools

5: Introduction to some more HTML elements

6: Introduction to SVG

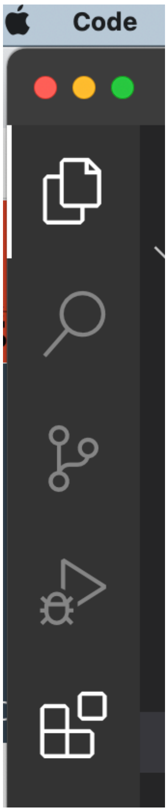
1: Install VS Code

Go to **https://code.visualstudio.com/ (https://code.visualstudio.com/)** and install this on your machines. Once installed, your tutors will demonstrate the basics of the software environment to help you get started. This is where you will do all of your coding work throughout the semester. There are excellent tutorials as well as a lot of help videos available on YouTube too - so often simply googling for help if you are stuck is a first great idea.

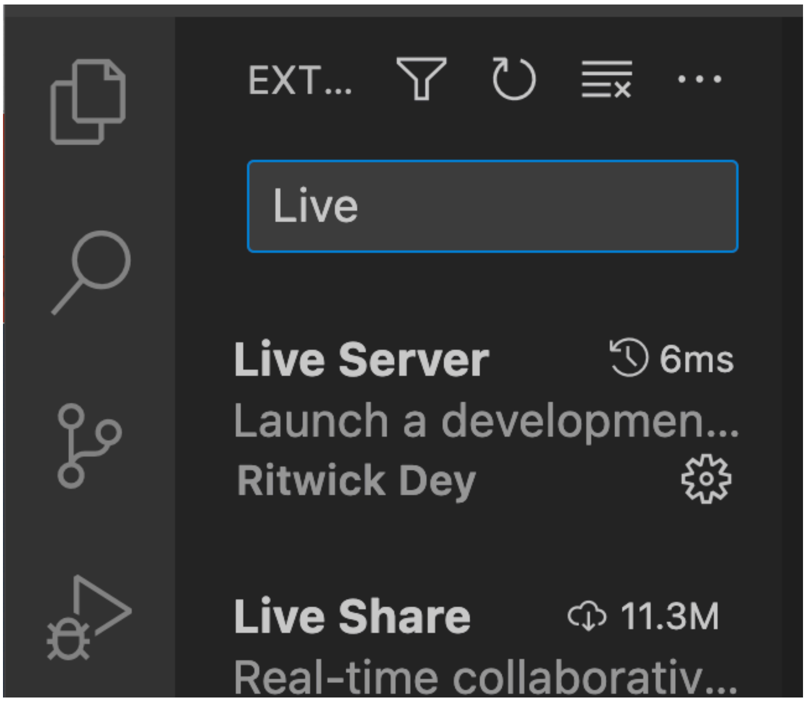


2: Install Live in VS Code

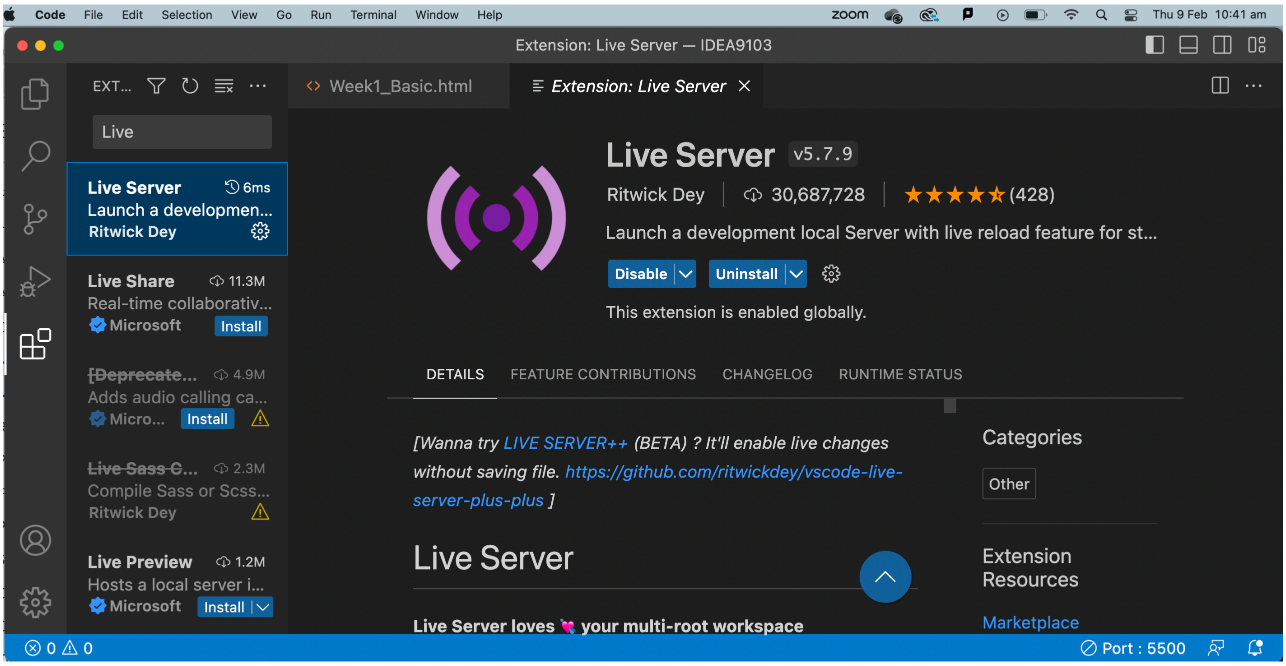
Fire up VS Code, and click on Extensions. On the left side bar, you see these icons, Extensions is the one at the bottom, highlighted bright white.



Search for Live, and an extension called Live Server will pop up as the first option.



Click on Live Server, and Install it.



Now shut down VS Code, and then restart it. When we write HTML pages, we will be able to run them via the Live Server.

Extension knowledge: The Client - Server Model

To understand why we installed this, you would need to understand the client-server model for the World Wide Web.

Do you know what happens when you type in a URL (Universal Resource Locater) or web "address" into a browser? How is a webpage loaded into your machine?

The answer is that the World Wide Web follows a client - server model. Read more about it here: **https://en.wikipedia.org/wiki/Client%E2%80%93server_model (https://en.wikipedia.org/wiki/Client%E2%80%93server_model)** and here: **https://en.wikipedia.org/wiki/Web_server (https://en.wikipedia.org/wiki/Web_server)**

The essence of this is that your local machine, the computer on which you "see" the webpage, is called a client. When you type in a URL into your browser, very roughly, the following happens:

- In your local machine, which is called the "client", you activate a communication request. This is you typing in a URL, which is like an address for a remote machine, called the "server", where your web page is actually sitting.
- This request reaches the server machine, and it finds the page you want, and sends it back to your client machine.
- The browser in your local machine receives this code and then renders the content, which you finally see as the webpage.

Now, modern browsers like Chrome have many safety features which prevent a webpage from running properly if this client-server model is not followed. Thus, even though we are running the webpages we are writing on our own machine, once we start using JavaScript, these pages will not render properly if we only double click on an HTML locally from our folders to open it in our browser. Thus, we need to mimic the client-server model, implying that we need to set up a server that can serve the webpages when our computer browser acts as a client.

What we have just done is install an extension called Live Server which enables us to do this.

3: Introduction to HTML

Start a new file in VSCode and type in the following (see below code) into it, save the file as Week1.html. Your tutors will again step you through what each of these elements mean. This content is also covered in detail in the lecture and the lecture slides, so if you missed the lecture, please have a listen to the lecture recording.

```
<!DOCTYPE html>
<html>
    <head>
        <title>IDEA9103: Week 1</title>
        <meta charset="UTF-8" />
    </head>
    <body>
        <p>Hello world!</p>
    </body>
</html>
```

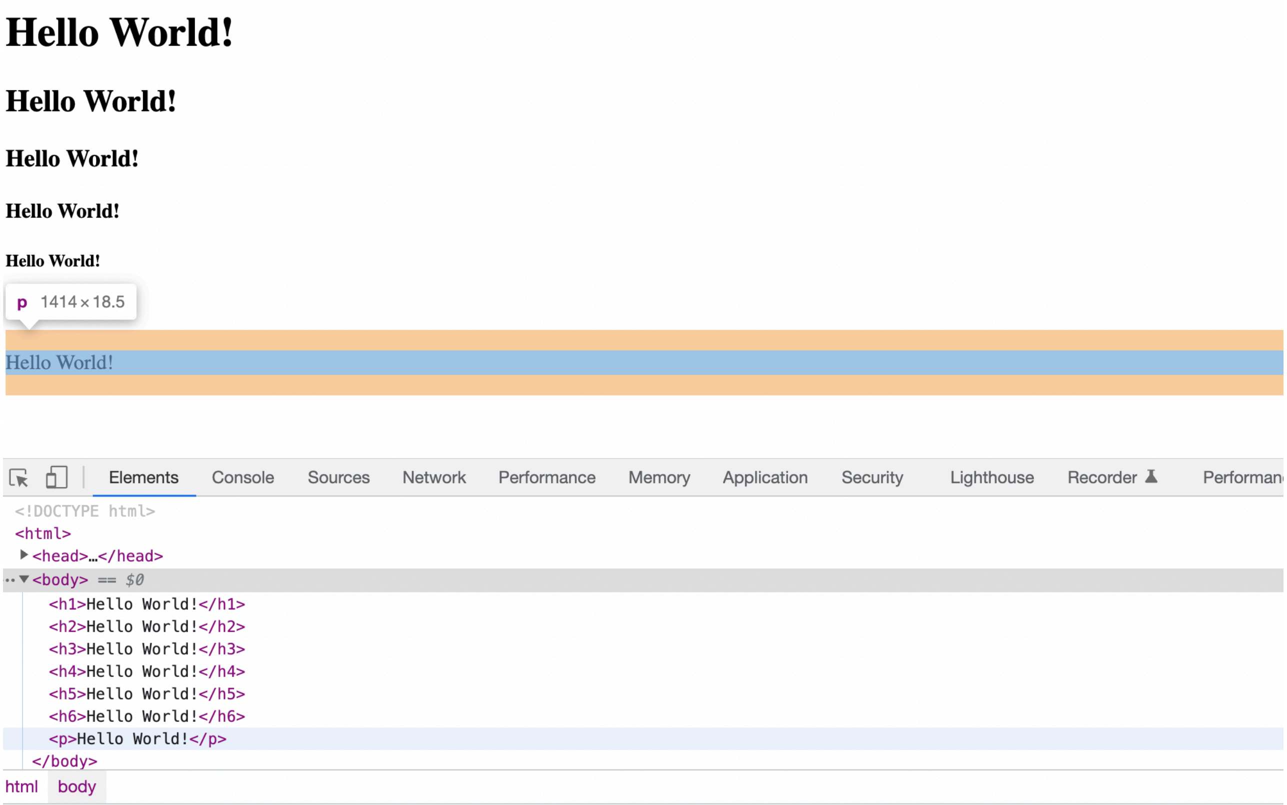
While saving files, we will be working with HTML, CSS and JS files. So, it would be a good idea to decide and then follow a folder structure for the whole semester. For example, your directory structure could look like: IDEA9103 --> Week 1 --> Week1.html for saving this particular file.

Save this file and run it in a browser. To run it, inside VS Code, right click on your saved file and choose "Open with Live Server". A new Chrome tab should open up where you can see the webpage running. Your tutors will show you how to do this.

4: Understand the Developer Tools

Your tutors will show you how to open Developer Tools in Chrome, and the Elements tab will show you the nested hierarchical Document Object Model structure of HTML pages. It is very important that you understand clearly how to use the Developer Tools - since it enables you to quickly inspect the structure of Webpages, as well as look at your JS code and its behaviour later when we get into JS programming.

For example, here is a snapshot showing how the paragraph element is highlighted on the main page if you hover over it in the code section of the Tools.



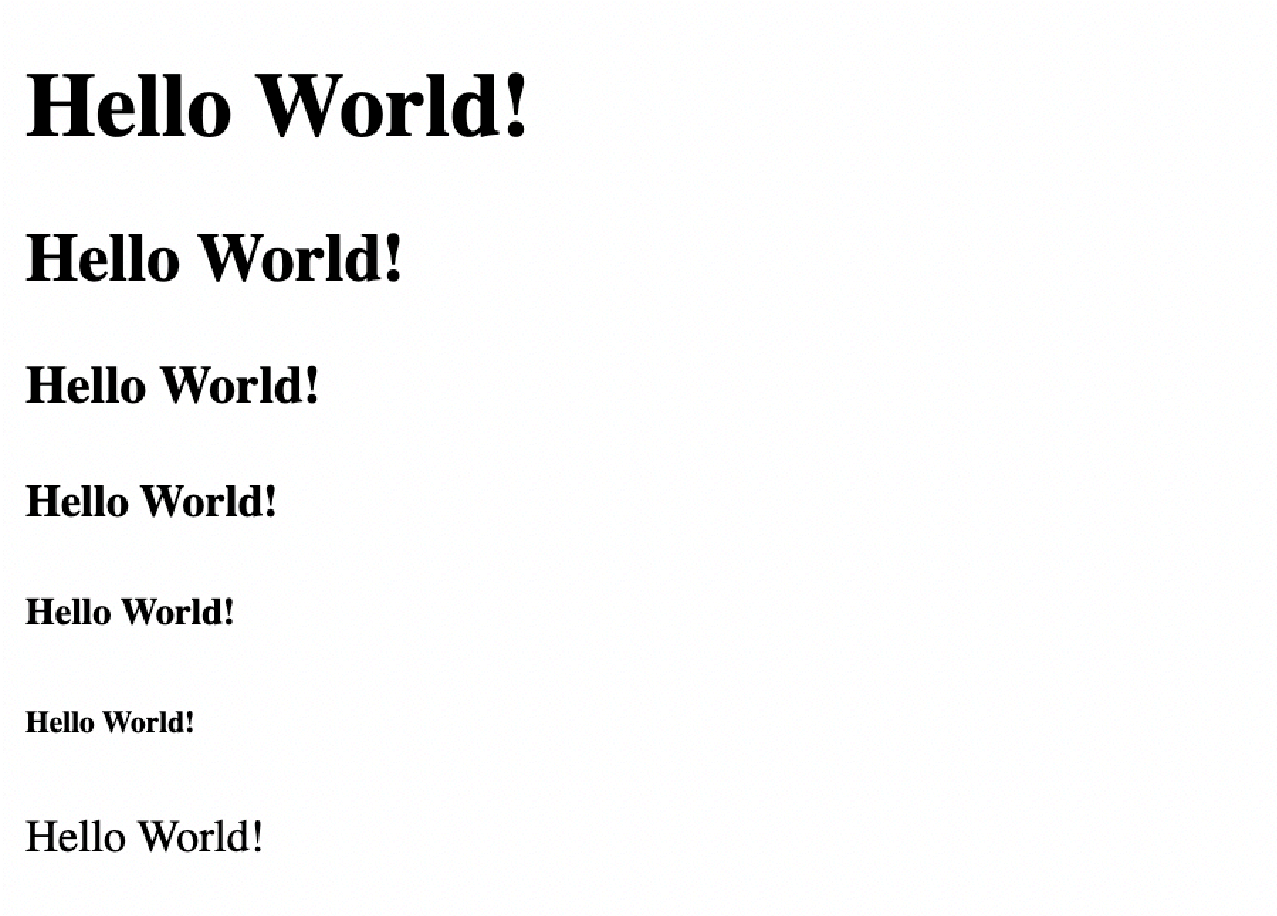
5: Introduction to more HTML elements

With help from your tutors, add in more code to your HTML page:

5.1 Headings

Headings in HTML are used to control the size of text which serves as titles or subtitles. They go from h1 (largest) to h6 (smallest). Each has its own tag. Try these out in your HTML file, adding them to the body, just above the <p> tag earlier, and then run the file to see how they render.

```
<!DOCTYPE html>
<html>
    <head>
        <title>IDEA9103: Week 1</title>
        <meta charset="UTF-8" />
    </head>
    <body>
        <h1>Hello world!</h1>
        <h2>Hello world!</h2>
        <h3>Hello world!</h3>
        <h4>Hello world!</h4>
        <h5>Hello world!</h5>
        <h6>Hello world!</h6>
        <p>Hello world!</p>
    </body>
</html>
```



5.2 The IMG tag

Add an image using the <img> tag

The <img> tag is used to embed images inside HTML pages. Download any image or use any image from your computer. I am using this image, you can also save it from here and use it.

Save it using the filename "helloworld.png". Type in the following code within the <body> tags:

```
<!DOCTYPE html>
<html>
    <head>
        <title>IDEA9103: Week 1</title>
        <meta charset="UTF-8" />
    </head>
    <body>
        <img src="helloworld.png" alt="Hello World" width="300" height="150">
        <p>Hello world!</p>
```

```
    </body>
</html>
```

There are a few things to note here.

- First, the <img> tag is a self-closing one, meaning a matching closing tag </img> is not needed.
- Second "src" means the source - the name of the image file. This needs to be stored in the same folder where your HTML file is, otherwise your code won't find the image file and it won't render.
- Third, the "alt" is like a name for your image, explaining what the image is about.
- Fourth, the "width" and the "height" specify, respectively, the width and the height of your image on the page.

5.3 Inline styling of elements

Styling any other HTML element using a similar form:

Did you note how the code above for the <img> is written or structured? This is a standard structure to write the set of attributes related to any element. Though we will find much more efficient and better ways of doing this, in principle, it is possible to style all of your individual elements by specifying their properties or attributes. You will see that even the more efficient ways follow a similar structure - just listing the attributes one after the other.

Just like the above code structure, you can also style any other HTML element by using a reserved word: "style" and then a list of style attributes separated by a semi-colon:

```
<!DOCTYPE html>

<html>
    <head>
        <title>IDEA9103: Week 1</title>
        <meta charset="UTF-8" />
    </head>
    <body>
        <h1 style="color:red; font-family:sans-serif">Heading 1</h1>
        <h2 style="color:blue; font-family:sans-serif">Heading 2</h2>
        <h3 style="color:orange">Heading 3</h3>
         <p style="color:green">Hello world!</p>
    </body>
</html>
```

> **Heading 1**
>
> **Heading 2**
>
> **Heading 3**
>
> Hello world!

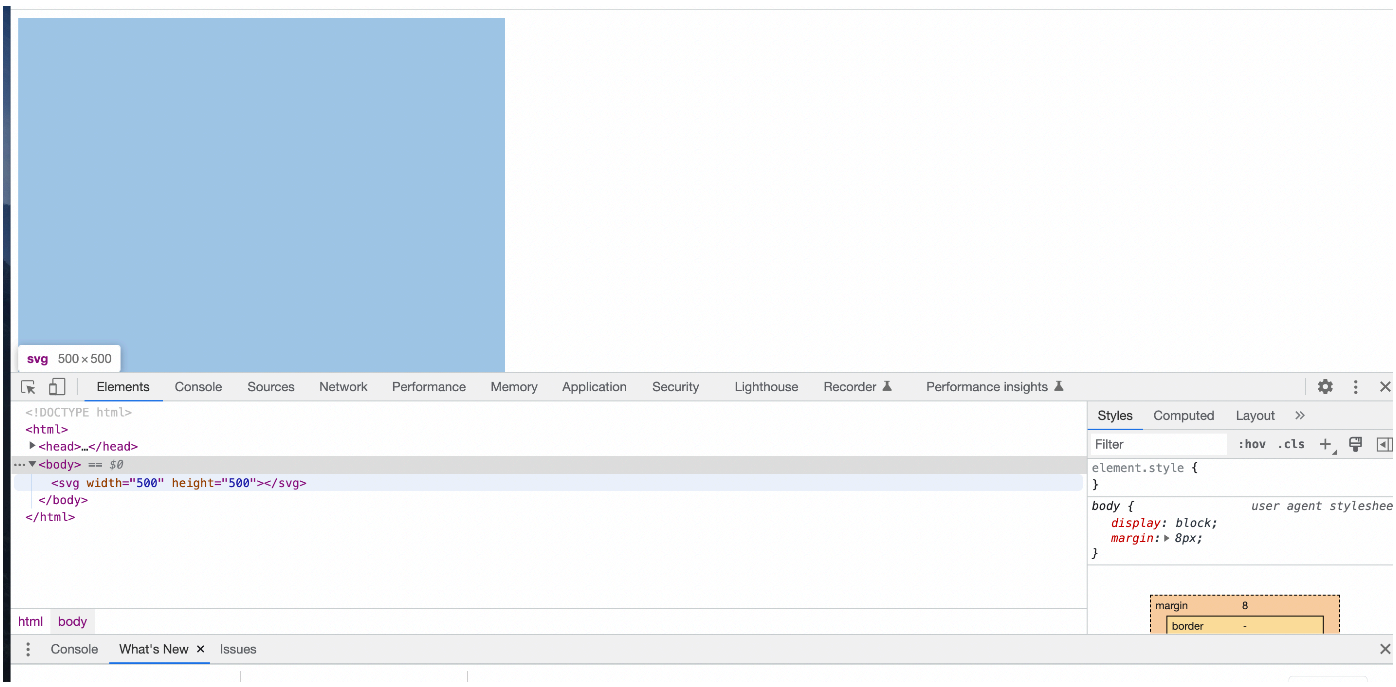
As you saw in the Lecture, we will normally use CSS files to style our content - what is shown here is BAD coding practice, you never style your HTML elements in this way. However, it is introduced here to show the general practice of separating different attributes and parameters by spaces and write them out as a list - whether you write CSS files, or whether you write SVG, as below.

6: Introduction to SVG

We are finally in a position to start drawing! As introduced in the lecture, SVG stands for Scalable Vector Graphics. You can use SVG just like normal HTML code to start drawing shapes, and render them with different colours, strokes, stroke-widths, etc. Your tutors will demonstrate the following steps. Go to **https://developer.mozilla.org/en-US/docs/Web/SVG (https://developer.mozilla.org/en-US/docs/Web/SVG)** to explore the SVG documentation. The SVG Element Reference Page shows you all of the possible things you can draw: **https://developer.mozilla.org/en-US/docs/Web/SVG/Element (https://developer.mozilla.org/en-US/docs/Web/SVG/Element)**. Today we will do basic drawing with simple shapes.

6.1 Add <svg> </svg> tags into your HTML page with a specified width and height

```
<!DOCTYPE html>
        <html>
            <head>
                <title>IDEA9103, Week 1</title>
                <meta charset="UTF-8" />
            </head>
            <body>
                <svg width="500" height="500">
                </svg>
            </body>
        </html>
```



If you inspect the code using the Developer Tools, hovering over the <svg> tag you can see that the HTML page has an SVG element rendered at the chosen width and height.

6.2 Add a circle

```
<!DOCTYPE html>
        <html>
```

```
        <head>
            <title>IDEA9103, Week 1</title>
            <meta charset="UTF-8" />
        </head>
        <body>
            <svg width="500" height="500">
                <circle
                cx="50"
                cy="50"
                r="30"
                stroke="black"
                stroke-width="4"
                fill="yellow">
                </circle>
            </svg>
        </body>
    </html>
```

The <circle> tag can be used to create circles, and the attributes of a circle can be specified in the same way. "cx" is the x-coordinate of the centre, "cy" is the y-coordinate of the centre. You will notice that in SVG the top left corner is (0,0), so the x- and y-coordinates are counted starting from the top left corner. "r" is the radius of the circle. The other attributes specify stroke, stroke-width and the fill colour.

6.3 Add a rectangle

Go to **https://developer.mozilla.org/en-US/docs/Web/SVG/Element/rect (https://developer.mozilla.org/en-US/docs/Web/SVG/Element/rect)** and follow a similar structure as above to create a rectangle.

6.4 Add a line

Go to **https://developer.mozilla.org/en-US/docs/Web/SVG/Element/line (https://developer.mozilla.org/en-US/docs/Web/SVG/Element/line)** and follow a similar structure as above to create lines.

6.5 Play with positions, colors, stroke widths, and other such attributes.

Now play around and change colors, strokes, stroke-widths and other attributes to generate multiple elements on a page. You will notice that the order in which you write the code decides how the elements show up. For example, adding a line followed by a circle will put the circle on top of the line. But if you flip the code and put the circle and then the line, then the line will show on top. Experiment writing some code using all of the knowledge learnt today, because this will help you attempt the Quiz task for today.

By the end of this step, you should know how to position these three types of SVG elements onto your HTML page. The quiz for this week asks you to create some artwork with SVG and submit an HTML page with your code. Once you are confidently working with the material in this tutorial, you can go ahead and attempt the quiz.

Happy coding!